# The Post-active ORBm and BMAp ensemble: what it is, and whether relapse needs it

Proposal for the next phase — two experiments on the two regions our whole-brain TRAP screen selected

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Morphine lick-PR / TRAP2;Ai14 / Xenium

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**WHAT I AM ASKING FOR**

1. Approval to run the Xenium pilot with Alina's lab in September, using the base mouse brain panel on tissue we already have.
2. Approval to order the custom probe panel in October once pilot QC passes.
3. Agreement on one design change to the chemogenetics plan: test at protracted abstinence rather than inside the 3-day Post window. Slides 8-9 show why the original plan would have produced a false negative.
4. Sign-off on group sizes. My first sketch (n=4) has ~29% power; I propose it as an expression pilot and a powered cohort behind it.

All timing parameters cited on the final slide.

STATUS

# Where we are

Three pieces of work are finished and feed directly into the next phase

### 1 WHOLE-BRAIN TRAP SCREEN (done)

20 mice, 10 Active / 10 Passive, terminal design: each mouse is sacrificed at one behavioural phase.

Phases sampled: During, Post, Withdrawal, Reinstatement.

Readout: tdTomato+ cell density per Allen region, Active vs Passive within phase.

### 2 REGION SELECTION (done)

7 candidate regions carried forward for spatial transcriptomics: ORBm, AId, CA, BMAp, LM, RE, CP.

Selection rule: Active > Passive in the craving phases (Post and Reinstatement) and still separated during Withdrawal.

**ORBm and BMAp are the two I propose to take forward now.**

### 3 PROBE PANEL (done)

Built from Allen Whole Mouse Brain 10X: per-subclass discriminating markers plus GPCR specificity tiers.

168 genes covering 20 Allen subclass populations across the two regions; 77 of them are the minimum that still identifies all 20.

Code and audit trail on GitHub.

**The screen has told us WHERE and WHEN. It cannot tell us WHAT those neurons are, and it cannot tell us whether the behaviour needs them. Those are the two experiments in this deck.**

Sources: TRAP_sample_manifest.csv; Fig06_7region_phase_delta.csv; Genelist_analysis_WMB v3.

# The result that drives the next phase

Active–Passive TRAP density separates at Post and Reinstatement — not while the mice are taking morphine

| Region | Cluster | Active – Passive (craving phases) | Take forward? |
|---|---|---|---|
| RE | 4 | 1.19 | |
| ORBm | 1 | 1.04 | YES — Experiment 1 + 2 |
| CA | 1 | 0.97 | |
| LM | 4 | 0.96 | |
| AId | 1 | 0.85 | |
| BMAp | 4 | 0.74 | YES — Experiment 1 + 2 |
| CP | 4 | 0.31 | |

### Where the Active–Passive difference appears across the 18-day paradigm

| FR 1-2 | Pre 3-5 | During 6-10 | Post 11-13 | Withdr. 14-16 | Re-exp 17-18 |
|---|---|---|---|---|---|
| | | | ▲ gap opens | ▲ gap narrows | ▲ gap returns |

morphine water fixed ratio

### What the number is

– Active minus Passive tdTomato+ density, z-scored within phase, averaged over Post and Reinstatement.

### What the phase pattern means

– The difference is not largest during morphine taking. It is largest at Post and again at Reinstatement, and it narrows during Withdrawal.

– So the ensemble that distinguishes a mouse that earned its morphine from a mouse that merely received it is engaged at Post and re-engaged at relapse.

### Why ORBm and BMAp specifically

– They come from the two different clusters, so together they sample a cortical and an amygdalar node rather than two versions of the same signal.

– Both have independent opioid literature: OFC is required for incubated oxycodone and heroin craving; amygdala ensembles are required for incubated craving across several drugs.

### Limitation I want to be explicit about

– This is a correlation from a terminal design with 4-6 mice per phase. It is a hypothesis generator, not a causal claim. That is what Experiment 2 is for.

Fig06_7region_phase_delta.csv; paired_slope_consistency.csv. Opioid literature cited on final slide.

PLAN

# Two questions, two experiments

Experiment 1 gives us the target. Experiment 2 gives us the causal claim.

**Q1 WHAT ARE THESE NEURONS? → EXPERIMENT 1: Xenium**

Cell-type identity of the Post-tagged ensemble in ORBm and BMAp, in situ, at single-cell resolution.

Which of the 20 Allen subclasses the tdTomato+ cells actually belong to — is the Active-specific signal one population or many?

Which druggable GPCRs those specific cells express.

Whether the Post ensemble and the relapse ensemble are the same cells (tdTomato and Fos co-detected in the same section).

**Output: a defensible molecular target, and the probe panel is already built.**

**Q2 IS THE BEHAVIOUR NECESSARY? → EXPERIMENT 2: TRAP-chemogenetics**

Silence the Post-tagged ensemble and ask whether morphine seeking falls.

Hypothesis: inhibiting the Post ensemble reduces seeking and relapse in Active mice, and does not in yoked Passive mice.

The Active-only prediction is the strong part of the hypothesis — it makes the experiment falsifiable rather than merely descriptive.

**Output: a causal statement about a specific ensemble in a specific region.**

**This experiment needs one design change from my first sketch. Slides 8-9.**

**Together they answer one sentence: the Post-active ensemble in region X consists of cell type Y expressing receptor Z, and relapse requires it. Either half alone is a weaker paper.**

# Experiment 1 — Xenium on the Post-tagged ensemble

One design choice does most of the work: tag at Post, read out at relapse

| | Specification | Why this and not something else |
|---|---|---|
| Mice | TRAP2;Ai14 (Fos2A-iCreER × Ai14)<br>JAX 030323 × 007914 | Ai14 is the reporter you have chosen; tdTomato mRNA is the TRAP tag Xenium reads. |
| Groups | Active n=3, Passive n=3 | Pilot scale. Statistics caveat on the next slide. |
| Tag | 4-OHT 50 mg/kg i.p. immediately after the behaviour day 12 session (Post day 2) | TRAP2 labels within ~6 h centred on the injection, so this captures the Post state. |
| Behaviour | Unchanged, days 1-18<br>(FR, Pre, During, Post, Withdrawal, Re-exposure) | Keeping the paradigm identical means the whole-brain TRAP result still maps onto it. |
| Abstinence | Days 19-31, home cage | Lets tdTomato accumulate and lets craving incubate. |
| Readout day | Day 32: cue-induced seeking test, sacrifice 90 min later | 20 days after tagging: well past your 10-day minimum, and Fos from the test is at peak. |
| Tissue | ORBm and BMAp coronal sections;<br>Active and Passive on the SAME slide | Puts the group comparison inside a slide instead of across slides. |
| Readouts | tdTomato (Post ensemble) + Fos (relapse ensemble) + 20 subclasses + GPCRs | The overlap of tdTomato and Fos is the single most informative measurement here. |

**Key point: tagging at Post and reading out at relapse turns one experiment into two measurements — what the Post ensemble is, and how much of it comes back at relapse.**

TRAP2 6-h window and 50 mg/kg 4-OHT: DeNardo et al. 2019 Nat Neurosci.

# Experiment 1 — the panel is built, with one procurement risk

168 genes in four blocks; the risk is the 100-gene cap on custom add-ons

| Block | What it answers | BMAp | ORBm | Union |
|---|---|---|---|---|
| 1 Cell-type unique markers | Which of the 20 populations is this cell? | 68 | 77 | — |
| 2 Cell-type-specific GPCRs | Druggable AND informative about cell type | 6 | 3 | — |
| 3 Broad GPCRs / shared markers | Druggable but group-level only | 29 | 29 | — |
| 4 IEG + reporter + backbone | Was it active, was it TRAPped, what class | 21 | 19 | — |
| TOTAL genes to order | | 124 | 128 | 168 |
| Minimum that still IDs all 20 | | 38 | 52 | 77 |

## BLOCK 4 IS THE TRAP READOUT — 4 GENES NEED CUSTOM PROBES

tdTomato and iCre are not mouse genes. Xenium supports exogenous targets, but 10x does not validate them, so these are at our risk.

tdTomato = the permanent tag (Ai14). iCre = the driver, which separates cells that expressed the driver from cells that completed recombination.

Fos, Npas4 and Arc give activity at the time of the test; Slc17a7 / Slc17a6 / Gad1 / Gad2 give the neurotransmitter class.

## PROCUREMENT RISK — RESOLVE BEFORE ORDERING

**A Xenium custom add-on is capped at 100 genes on top of a pre-designed base panel. Our list is 168.**

Option A (preferred): Xenium Prime 5K mouse panel as the base. It should already contain nearly all 164 endogenous genes, leaving only the 4 transgenes as custom.

Option B: base mouse brain panel + 100-gene add-on. Feasible because the 77-gene minimum panel plus the transgenes fits inside the cap. This is exactly why I tiered the list.

**Action: cross-check our 168 genes against the base panel gene lists and get a quote for both options. Two days of work, must happen before the October order.**

Statistics caveat: n=3 per group means the unit of inference is the mouse, not the cell. I will analyse cells nested within mouse and report mouse-level effects. Treat Experiment 1 as descriptive; if QC passes I would extend to n=5 per group.

# Experiment 1 — schedule

The September pilot deliberately uses the base panel only, so tissue QC is not blocked by panel design

| When | Step | What must be true to proceed |
|---|---|---|
| Sep 2026 | Xenium pilot with Alina's lab. Base mouse brain panel only, 1 Active + 1 Passive brain from existing tissue. | Nothing — this is the gate, and it needs no new panel. |
| Sep 2026 | QC criteria fixed in advance: transcripts per cell, negative-probe rate, segmentation quality, registration of ORBm and BMAp to Allen, and recovery of the expected subclasses. | Agree the numeric thresholds with Alina BEFORE the run, so pass/fail is not a judgement call. |
| Sep-Oct 2026 | Cross-check the 168 genes against base panel content; quote Prime 5K vs base + 100 add-on. | Pilot QC passed. |
| Oct 2026 | Submit panel design, including the 4 transgene FASTA sequences (tdTomato, iCre, and EGFP/WPRE if wanted). | Budget option chosen; Ai14 and TRAP2 construct sequences in hand. |
| Oct-Nov 2026 | Panel manufacture and delivery (allow 6-8 weeks). | Order placed. |
| Nov 2026 | Run the behaviour cohort for Experiment 1 (n=3+3) so tissue is ready when the panel arrives. | Mice genotyped TRAP2;Ai14; 4-OHT prepared fresh. |
| Dec 2026 - Jan 2027 | Xenium run on ORBm and BMAp; analysis. | Panel delivered; sections cut and QC'd. |

**The one thing that can slip this schedule is the panel cross-check. If we discover in October that our genes do not fit the cap, we lose the manufacturing slot.**

# Experiment 2 — the problem with my first sketch

Injecting CNO on Post day 7 would have tested nothing, because the receptor is not there yet

## TRAP IS INTRINSICALLY RETROSPECTIVE

**To be tagged, a neuron must be active. To be silenced, it must first manufacture the receptor. Those cannot happen at the same time.**

So TRAP can never answer 'does activity during Post cause behaviour during Post'. It answers 'are the neurons that were active during Post required for behaviour later'.

This is a property of the method, not a flaw in the plan — but it decides what the experiment is allowed to claim.

## AND THE ARITHMETIC DOES NOT WORK

**AAV-DIO-hM4Di needs about 14 days after 4-OHT before hM4Di is functional, on top of ~3 weeks for the AAV itself before tagging.**

My sketch: 4-OHT on Post day 2 (behaviour day 12), CNO on 'Post day 7' — five days later.

Five days is not enough. We would inhibit nothing, see no behavioural effect, and be unable to tell that apart from the hypothesis being wrong.

| Requirement | Needs | My sketch gave it | Verdict |
|---|---|---|---|
| AAV established before tagging | ~21 days | not specified | fixable |
| hM4Di functional after 4-OHT | ~14 days | 5 days | FAILS |
| Post phase long enough to test inside it | ~17 days | 3 days (day 11-13) | FAILS |

**A null result from an underpowered, under-expressed experiment is the worst outcome available to us: it costs a full cohort and answers nothing. The next slide is the fix.**

14-day post-4-OHT window for AAV-delivered DREADDs: TRAP2 ghrelin study, Int J Mol Sci 2022. 3-4 weeks for Cre-dependent AAV expression is standard.

# Experiment 2 — the constraint points at the right experiment

Three independent timescales all land on the same window, about two weeks after the last morphine

| Timescale | Value | Source |
|---|---|---|
| hM4Di becomes functional after 4-OHT | ~14 days | TRAP2 + AAV-DIO-DREADD studies |
| Incubation of craving peaks | ~withdrawal day 14 | Fos ensemble studies, inverted-U across drugs |
| OFC inactivation reduces opioid seeking | at day 15, NOT at day 1 | Oxycodone incubation, OFC inactivation |

**The technique is ready exactly when the behaviour is maximal, and exactly when the region is known to matter. The delay we were fighting is the delay we want.**

## WHAT CHANGES, AND WHAT DOES NOT

**Does NOT change: behaviour days 1-18 stay exactly as run. Same FR, Pre, During, Post, Withdrawal, Re-exposure. The whole-brain TRAP and Xenium data still map on.**

**Changes: we add a home-cage abstinence period and a drug test at the end. We add days after day 18; we do not alter days 1-18.**

That is the whole argument for this redesign — it is additive, not a new paradigm.

## THE QUESTION, RESTATED HONESTLY

**Cannot test: does Post activity cause Post seeking?**

Can test, and this was the second half of my own hypothesis:

**Are ORBm / BMAp neurons that were active during Post necessary for morphine seeking after protracted abstinence, and is that requirement specific to mice that earned the drug?**

This is also the more clinically relevant question. Patients do not relapse during use; they relapse after abstinence.

## IF YOU WANT THE WITHIN-POST TEST ANYWAY

There is a way to do it at 48 h instead of 14 days. Slide 13.

Full citations on the final slide.

# Experiment 2 — design

2 × 2 between subjects, drug crossed within subject; one region at a time

| | Levels | Purpose |
|---|---|---|
| Group (between) | Active / yoked Passive | The Active-only prediction. Passive is the specificity control. |
| Virus (between) | AAV-DIO-hM4Di-mCherry / AAV-DIO-mCherry | The mCherry + drug arm is the essential control: it removes any effect of the ligand itself. More important than a saline arm. |
| Drug (WITHIN subject) | DCZ 0.1 mg/kg / vehicle, crossover | Each mouse is its own control, which roughly halves the mice needed. |
| Region | ORBm first, then BMAp | ORBm has the larger Active−Passive difference and is the easier bilateral target. BMAp second, informed by the Xenium result. |

Starting coordinates to verify in cohort 1, then fix: ORBm ~ AP +2.6, ML ±0.35, DV −2.7; BMAp ~ AP −1.8, ML ±2.4, DV −5.1. Use 150-250 nl at low rate — BMAp sits next to MEA, BLA and the sAMY-like GABA populations our marker work flagged, so spread must be verified histologically.

| n per cell | dz=0.8 | dz=1.0 | dz=1.2 | dz=1.5 |
|---|---|---|---|---|
| 4 (my sketch) | 21% | 29% | 38% | 53% |
| 8 | 50% | 68% | 83% | 95% |
| 10 | 62% | 80% | 92% | 99% |
| 12 | 71% | 88% | 97% | 100% |

### Power, paired t-test, two-tailed α=0.05

– Reported chemogenetic effects on drug seeking are large, roughly a 40-50% reduction, so dz ~ 1.0-1.2 is a fair planning assumption.

– n=4 detects nothing smaller than dz~1.6. n=10 is the first row where a large effect is reliably detectable.

### What I propose instead of choosing one n

– Cohort 1 = expression pilot, n=4 per cell (16 mice, ORBm). No behavioural inference claimed. It buys the AAV coordinates, the percentage of tdTomato+ cells that become hM4Di+, confirmation that DCZ suppresses Fos in those cells, and a variance estimate.

– Cohort 2 = powered, n=10 per cell (40 mice, ORBm), with n set by cohort 1's variance rather than by my guess.

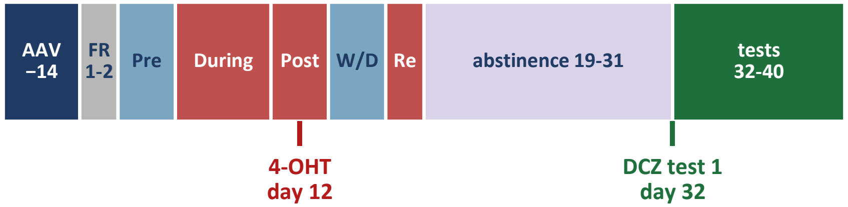
– Allow ~25% attrition for missed targeting and health, so order accordingly.

# Experiment 2 — day-by-day schedule

Behaviour days 1-18 are unchanged; everything new is appended after day 18

| Day | Phase / step | Reward | Note |
|---|---|---|---|
| −14 | Bilateral AAV surgery (hM4Di or mCherry), ORBm | — | 26 days before tagging, comfortably past the ~21-day AAV requirement |
| −13 to 0 | Recovery, handling, injection habituation | — | Handling matters: it lowers TRAP background labelling |
| 1-2 | Fixed ratio | — | As currently run |
| 3-5 | Pre | Water | As currently run |
| 6-10 | During | Morphine | Passive lick lockout applies, as currently run |
| 11-13 | Post | Morphine | 4-OHT 50 mg/kg i.p. right after the DAY 12 session |
| 14-16 | Withdrawal | Water | As currently run |
| 17-18 | Re-exposure | Morphine | As currently run. Day 18 = last morphine |
| 19-31 | Forced abstinence, home cage | — | hM4Di matures (19 days post-4-OHT by day 31) and craving incubates |
| 32 | TEST 1 — cue-induced seeking, no morphine | — | DCZ or vehicle 15 min before. Withdrawal day 14. Primary endpoint |
| 34 | TEST 2 — same test, treatment crossed over | — | 48 h washout is ample for DCZ |
| 36 | SPECIFICITY — water or sucrose PR under DCZ | Water | Shows the effect is not general motivation |
| 38 | MOTOR — locomotion / open field under DCZ | — | Rules out a movement confound |
| 40 | Final session, sacrifice 90 min later | — | Histology: placement, tdTomato ∩ hM4Di overlap, Fos suppression |

## Same picture as one timeline




### THE 20-DAY GAP IS THE POINT

**Day 12 (4-OHT) to day 32 (first test) = 20 days.**

hM4Di needs ~14 days after tagging → satisfied with 6 days of margin.

Your own requirement of at least 10 days after TRAP → satisfied.

Day 32 is withdrawal day 14 from the last morphine, which is where incubated opioid seeking peaks and where OFC inactivation is known to work.

**Days 1-18 are untouched, so the whole-brain TRAP and Xenium results still describe this cohort.**

morphine water abstinence test

Primary endpoints: active licks and PR breakpoint in the 1-h seeking test. Analysis: Group × Virus × Drug mixed-effects model with mouse as random intercept.

EXPERIMENT 2

# Experiment 2 — the controls a reviewer will ask for

Each one closes a specific alternative explanation

| Control | What it rules out | Cost |
| --- | --- | --- |
| AAV-DIO-mCherry + DCZ | That the ligand itself changed behaviour. This is the control that matters most and it is why I am not relying on a saline arm. | Built into the 2×2 |
| Vehicle in hM4Di mice (crossover) | Baseline drift and order effects. | Free — within subject |
| No-4-OHT (oil vehicle) + AAV | Cre-independent recombination leak, i.e. hM4Di in cells that were never active. | 3-4 mice, histology only |
| Home-cage 4-OHT, no session | That we tagged the paradigm rather than the Post state. Gives the background labelling rate to subtract. | 3-4 mice, histology only |
| Water or sucrose PR under DCZ | A general drop in motivation or in licking ability. | One session, same mice |
| Locomotion under DCZ | A motor confound. | One session, same mice |
| Fos in hM4Di+ cells after DCZ | That we inhibited nothing. This is target engagement and without it a null result is uninterpretable. | Terminal histology |

### USE DCZ, NOT CNO

**CNO back-converts to clozapine in vivo, and clozapine crosses the blood-brain barrier and alters motivation on its own.**

In a drug-seeking assay that confound sits directly on top of the dependent variable, so it is not a risk we should take.

DCZ: higher affinity, faster brain entry, no detectable activity across 318 off-target GPCRs. Effective at 0.1 mg/kg i.p. or lower.

### OPEN QUESTION I NEED HELP WITH

The TRAP2 protocol single-houses mice for 48 h before tagging, to keep background labelling low.

Our yoked design pairs each Active with a Passive partner. If yoking requires them to be co-housed on day 12, we cannot single-house them.

**Fallback: keep normal housing, treat both groups identically, and measure the background with the home-cage 4-OHT control rather than trying to eliminate it.**

CNO reverse metabolism: Gomez et al. 2017 Science; Manvich et al. 2018 Sci Rep. DCZ: Nagai et al. 2020 Nat Neurosci.

# If you want inhibition inside the Post window

There is a clean way to do it, but it needs new breeding — so I propose it as phase 2

| | A — Protracted abstinence (RECOMMENDED, start Nov) | B — Transgenic hM4Di + cannula (phase 2) | C — Extended Post (not recommended) |
|---|---|---|---|
| Effector | AAV-DIO-hM4Di into ORBm / BMAp | TRAP2 × R26-LSL-hM4Di (JAX 026219). Effector already in the genome, so no AAV lag | AAV-DIO-hM4Di |
| Delay from 4-OHT to test | ~20 days | 48 h | ~15 days, but inside an extended Post |
| Region specificity | From the injection site | From local DCZ infusion through cannulae — the transgene is brain-wide | From the injection site |
| What it tests | Is the Post ensemble needed for relapse after abstinence? | Is the Post ensemble needed for seeking within the Post phase? | Same as B, but in a paradigm we have not characterised |
| Main cost | None beyond the appended days | New cross; Rosa26 conflict means this line cannot also carry Ai14, so the tag becomes mCitrine. Cannula surgery. 4-6 months of breeding | 3 extra weeks of morphine per mouse, so drug history no longer matches the 18-day model behind our TRAP data |
| Ready to start | November 2026 | Mid 2027 if we start breeding now | November 2026 |

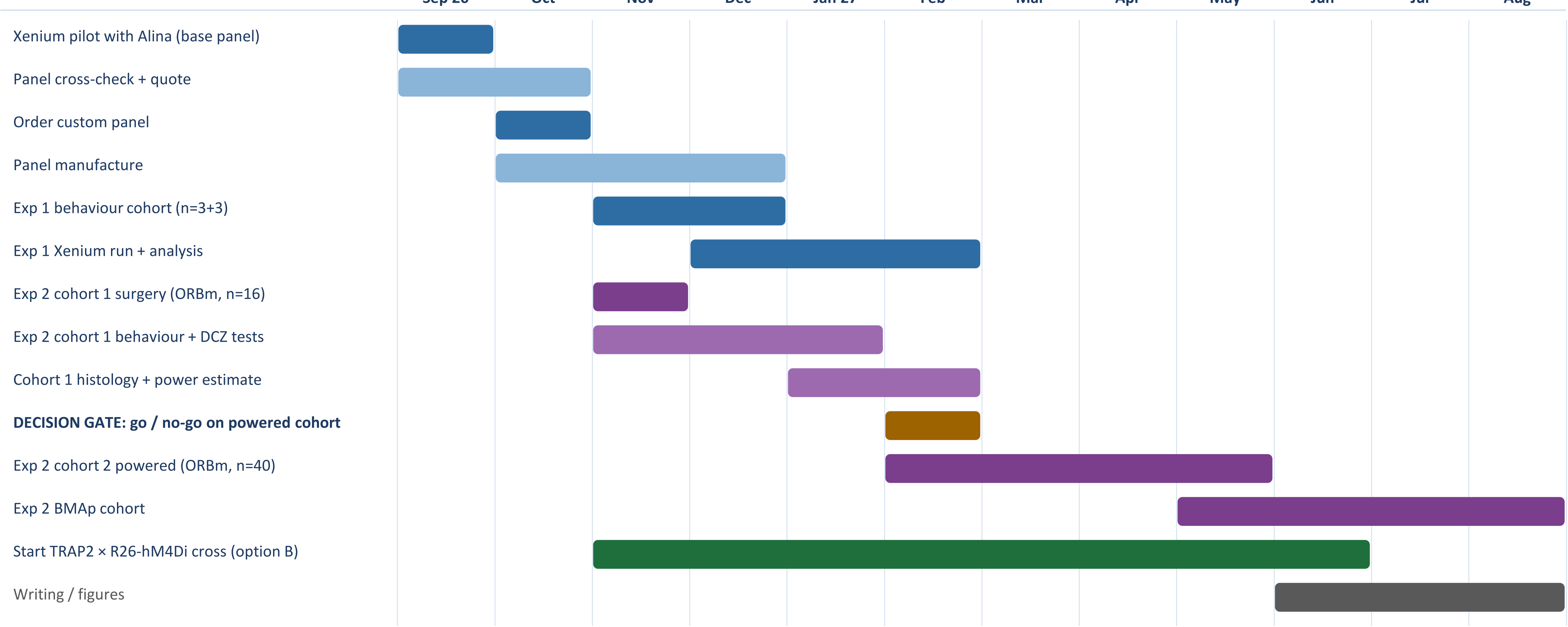
**My recommendation: run A now, and start the B cross in parallel so the within-Post question is answerable next year. A and B together are a much stronger story than either alone, because they separate 'needed while taking' from 'needed for relapse'.**

48-h transgenic hM4Di precedent with intracerebral CNO: LEC engram study, bioRxiv 2025. R26-LSL-hM4Di: JAX 026219, Rosa26 locus — same locus as Ai14 (JAX 007914).

SCHEDULE

# Master timeline

September 2026 to August 2027




**The decision gate in February is deliberate: no powered cohort is ordered until cohort 1 has shown that hM4Di reaches the tagged cells and that DCZ suppresses their activity.**

# What I need from you, and what could still go wrong

## DECISIONS I NEED

1. Budget route for the panel: Xenium Prime 5K base, or base panel plus a 100-gene add-on. This changes what I can order in October.
2. Which region first for chemogenetics. I recommend ORBm — larger effect in the screen, easier bilateral target, and cortex is more forgiving of spread.
3. Group sizes: do you accept cohort 1 as an explicitly underpowered expression pilot, with the powered cohort gated on its result?
4. Housing on the tagging day — whether the yoked design can tolerate 48 h of single housing before day 12, or whether we measure background instead.
5. Sex as a variable. The current cohort needs checking; if it is male-only I would like to add females to the powered cohort rather than to the pilot.
6. Whether to start the TRAP2 × R26-hM4Di cross now, so the within-Post question is answerable in 2027.

## RISKS, AND WHAT I WOULD DO

Custom transgene probes are not validated by 10x. If tdTomato detection is weak, iCre gives a second independent readout of the same cells, and immunostaining on adjacent sections is the fallback.

BMAp is small and next to MEA, BLA and the sAMY-like GABA populations. Mitigation: small injection volume, and every mouse verified histologically against the subclass markers from the panel work.

Our Post ensemble may not be one cell type. If Xenium shows it is heterogeneous, the chemogenetic result becomes harder to interpret — which is a reason to let Experiment 1 report before the powered cohort is locked.

**Morphine dose and delivery volume are not written down anywhere I can find in the code or manifests. I need them for the methods section and for the licence.**

Oral lick-based self-administration is less standard than IV, so reviewers will ask for intake per session in mg/kg. Worth agreeing how we report that now.

# References for every parameter used in this deck

| Claim in this deck | Source |
|---|---|
| TRAP2 labels within a ~6 h window centred on 4-OHT; 4-OHT 50 mg/kg i.p. at 10 mg/mL in 1:4 castor:sunflower oil; single housing 48 h before tagging | DeNardo LA et al. Temporal evolution of cortical ensembles promoting remote memory retrieval. Nat Neurosci 2019;22:460-469 |
| TRAP2 = Fos2A-iCreER, JAX 030323;  Ai14 = JAX 007914 | DeNardo et al. 2019; Madisen L et al. Nat Neurosci 2010;13:133-140 |
| AAV-delivered Cre-dependent DREADDs need ~2 weeks after 4-OHT, on top of ~3 weeks for AAV expression before tagging | López-Ferreras L et al. TRAPing ghrelin-activated circuits. Int J Mol Sci 2022;23:559; Cre-dependent AAV DREADD studies use 3-4 week expression windows |
| Transgenic Cre-dependent hM4Di can be used 48 h after tagging, with local intracerebral ligand infusion for region specificity | LEC engram cell study, bioRxiv 2025 (TRAP2 × hM4Di-DREADD, 48 h, intra-LEC CNO) |
| R26-LSL-hM4Di/mCitrine = JAX 026219, targeted to the Rosa26 locus (hence the conflict with Ai14) | Zhu H et al., Cre-dependent DREADD mice; JAX strain 026219; MGI:5614050 |
| CNO is reverse-metabolised to clozapine and produces clozapine-like effects | Gomez JL et al. Science 2017;357:503-507; Manvich DF et al. Sci Rep 2018;8:3840 |
| DCZ is more potent and selective than CNO; effective at 0.001-0.1 mg/kg with no activity across 318 off-target GPCRs | Nagai Y et al. Nat Neurosci 2020;23:1157-1167; Nentwig TB et al. Sci Rep 2022;12:6520 |
| OFC inactivation reduces oxycodone seeking on withdrawal day 15 but not day 1 | Fredriksson I / Shaham lab. Role of orbitofrontal cortex in incubation of oxycodone craving. Addict Biol 2020;25:e12927 |
| Fos ensembles are causally required for opioid and nicotine seeking; incubation peaks around withdrawal day 14 | Funk D et al. J Neurosci 2016 (CeA, nicotine); Fredriksson I et al. Sci Adv 2023 (vSub, oxycodone); Warren BL et al. Addict Biol 2022 (IL, oxycodone) |
| Xenium supports exogenous targets (reporters, transgenes) via advanced custom design from a FASTA of ≥80 bp, but 10x does not validate them; add-on panels are capped at 100 genes | 10x Genomics CG000683 Xenium Custom Panel Design; CG000643 Designing Custom Xenium Panels |